

ONE-RELATOR GROUPS

See also problems (04), (05), (06) in the [OUTSTANDING PROBLEMS](#) section.

*(OR1) (G.Baumslag) Are all one-relator groups with torsion residually finite? [Background](#)

(OR2) Is the isomorphism problem solvable for one-relator groups with torsion? [Background](#)

(OR3) (A.Miasnikov) Is the complexity of the word problem for every one-relator group quadratic, i.e., is there for every one-relator group an algorithm solving the word problem in quadratic time with respect to the length of a word? In polynomial time?

(OR4) Is the subgroup membership problem solvable for one-relator groups? That is, is there an algorithm for deciding if a given element of the group belongs to a given finitely generated subgroup? [Background](#)

(OR5) Is it true that if the relation module of a group G is cyclic, then G is a one-relator group? [Background](#)

(OR6) (G.Baumslag) Let $H = F/R$ be a one-relator group, where R is the normal closure of an element r in F . Then, let $G = F/S$ be another one-relator group, where S is the normal closure of $s = r^k$ for some integer k . Is G residually finite whenever H is?

(OR7) (G.Baumslag) Let $G = F/R$ be a one-relator group with the relator from $[F, F]$.

(a) Is G hopfian?

*(b) Is G residually finite?

(c) Is G automatic? [Background](#)

(OR8) (G.Baumslag) The same as (OR7), but for a relator of the form $[u, v]$. [Background](#)

*(OR9) (D.Moldavanskii) Are two one-relator groups isomorphic if each of them is a homomorphic image of the other? [Background](#)

(OR10) Is every one-relator group without metabelian subgroups automatic? [Background](#)

(OR11) (C.Y.Tang) Are all one-relator groups with torsion conjugacy separable?

(OR12) Are all freely indecomposable one-relator groups with torsion co-hopfian?

(OR13) (a) Which finitely generated one-relator groups have all generating systems (of minimal cardinality) Nielsen equivalent to each other?

(b) Which finitely generated one-relator groups have only tame automorphisms (i.e., automorphisms induced by automorphisms of the ambient free group)?

(OR14) (G.Baumslag) Describe one-relator groups which are discriminated by a free group. [Background](#)

(OR15) (B.Fine) If G is a one-relator group with the property that every subgroup of finite index is again a one-relator group, and every subgroup of infinite index is free, must G be a surface group?

*(OR16) Let S_n be the orientable surface group of genus n .

(a) Are the groups S_n and S_m ($m, n > 1$) elementarily equivalent? (i.e., $\text{Th}(S_m) = \text{Th}(S_n)$)?

(b) Is S_m elementarily equivalent to F_{2m} , the free group of rank $2m$? [Background](#)